University of Victoria
Department of Physics and Astronomy

Arif Babul
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April 15th, 2026

Re: Letter of Collaboration

I am pleased to confirm my collaboration with Prof. Maarit Korpi-Lagg and their team, headquartered at the HPC lab in the department of Computer Science at Aalto University, on investigating dust processing in a turbulent interstellar medium (ISM) using the extended multi-scale code-coupling framework.

This project addresses an important tension in astrophysics: the observed dust abundances in starburst galaxies do not agree with current theoretical predictions. The latter are based on the expectation that SN blast waves ought to be very efficient at destroying the dust. Intriguingly, recent results from high resolution but small-scale simulations of the ISM by the present team suggest that the dust destruction rate in the ISM is overestimated and that the actual destruction rate turns out to be significantly lower when one accounts for the turbulence and magnetic fields in the ISM. In these simulations, the magnetic field is the result of small-scale dynamos and the resultant fields are weaker than observed. The expectation is that the weak field generated by small-scale dynamos is almost certainly augmented by magnetic fields on larger scales that are observed in real galaxies. The latter are thought to arise self-consistently via large-scale dynamo effects. Modelling the large-scale magnetic field self-consistently, however, would require simulations with larger domains evolved over gigayears and including the microphysics of dust processing in these larger simulations is, from a computational perspective, prohibitively expensive. Therein lies the challenge: how can one jointly model the small-scale dust microphysics and capture the relevant larger-scale physical processes in a consistent fashion?

The proposed research programme aims to do this via a multi-scale simulation framework where simulations with successive smaller box sizes (and hence, higher spatial/temporal resolution) essentially zoom in on a small volume within the large-scale simulation. In this framework, the successive smaller volumes inherit large-scale conditions from their parent box. By coupling the simulations in this manner, it is possible to realistically model the physical processes across the full hierarchy of relevant physical and temporal scales and achieve a more complete and physically consistent picture of dust evolution in the interstellar medium.

Apart from leveraging the multi-scale simulation framework, the programme will also leverage advanced machine learning and artificial intelligence methods to enhance the efficiency and the scientific reach of the simulations. These tools will enable automated feature detection, adaptive selection of regions of interest, and intelligent management of computational resources, facilitating real-time data analysis and more effective coupling between simulations across the scales.

The proposed program is timely, high impact, and scientifically very important. JWST's observations are leading to unprecedentedly detailed maps of the 3D structure of the interstellar dust and gas in different galactic environments. The proposed programme has the potential to substantially advance our understanding of how these structures and features form. At the same time, JWST observations are indicating a greater than anticipated abundance of dust-obscured galaxies in the early universe, with many of these being galaxies with high star formation rates. Current models of galaxy evolution cannot account for these observations because – as already noted – of the inherent assumption that SN blast waves ought to be very efficient at destroying the dust, an assumption that may not hold under more realistic conditions.

I am specifically interested in the revised understanding of dust evolution that this programme promises, and its potential impact on our models of galaxy evolution. My research interest (and expertise) is in modeling galaxy evolution using cosmological hydrodynamic numerical simulations and I am convinced that improving the dust modeling in these simulations is key to mitigating the current tension between theory and observations. Moreover, if the dust can survive the SN blast waves and can be successfully expelled into the circumgalactic medium, it can have a significant impact on the latter's cooling profile and that in turn, would further impact galaxy evolution.

To conclude, I would like to, once again, confirm my strong interest in this project and my commitment to contribute my experience and expertise to this initiative. I commit to contributing actively to the scientific discussions of the results as well as their wider implications, and to ensuring that the results are disseminated widely within the astrophysics community

Yours sincerely,

Arif Babul
Fellow, American Physical Society
University of Victoria Distinguished Professor
Leverhulme Visiting Professor @ Univ. of Edinburgh

American Museum of Natural History

Mordecai-Mark Mac Low
Curator & Professor

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29 April 2026

Review Panel
International Collaboration in High-Performance Computing
Academy of Finland

Dear Colleagues:

I am enthusiastic to collaborate with Maarit Korpi-Lagg and Frederick Gent on their project entitled "Extended multiscale code coupling for dust processing in supernova-driven turbulence" submitted in response to the 2026 call.

I am currently a tenured professor and curator in the Department of Astrophysics at the Gilder Graduate School of the American Museum of Natural History, as well as an adjunct professor in the Department of Astronomy of Columbia University, both in the United States. I am the PI on a US Department of Energy INCITE grant with Gent as coI that is being used to run Pencil Code-Astaroth on the Frontier supercomputer, the largest publicly available machine in the country. I have published multiple papers with the PI and coI on supernova-driven dynamos in the interstellar medium (Gent et al. 2020, 2021, 2023, 2024, 2025, 2026) as well as a review article in Living Reviews of Computational Astrophysics (Korpi-Lagg et al. 2024). I have also published on dust interactions with supernova remnants (Slavin et al. 2020) as well as extensively on the properties of the supernova-driven interstellar medium (e.g. Mac Low et al. 2001, 2005, de Avillez et al. 2002, Joung et al. 2006, Hill et al. 2012, 2018, Gatto et al. 2015). My research has been recognized with a Humboldt Research Prize and Fellowship in the American Astronomical Society.

With this letter I commit to fulfilling my responsibilities in the proposed collaboration of advising on interstellar medium and dust properties and processes, assisting in formulating science goals and interpreting results, and collaborating on publications. Our successful collaboration to date has been primarily implemented using weekly online video conversations as well as shared Git repositories for code and publications. I will suggest that the graduate student to be supported by this grant visit my institution to focus on interstellar medium properties and processes. I have found short visits of a month or so to be quite effective in past collaborations.

I am of course happy to provide any additional information that may be useful.

Yours,

Mordecai-Mark Mac Low

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Stockholm, May 2, 2026

To the Academy of Finland
Re: Letter of Collaboration in support of the application by Maarit Korpi-Lagg

Dear Members of the Review Committee,

I am writing to express my strong support for the funding application submitted by Maarit Korpi-Lagg and Frederick Gent concerning their proposed research on dust modelling. I am highly enthusiastic about this project, which represents a natural and timely continuation and significant expansion of a research programme that I initiated over a decade ago. Should the Academy of Finland decide to fund this proposal, I will be fully committed to contributing as an active collaborator.

Below, I outline the nature of my involvement in accordance with the Academy's guidelines.

Collaborator Information

- **Name:** Lars Mattsson
- **Position:** Assistant Professor in Astrophysics
- **Affiliation:** Nordita, Stockholm University

Purpose, Objectives, and Added Value of the Collaboration

The primary aim of my participation is to contribute the expertise and experience I have developed over the past decade as principal investigator of the original project (initiated in 2016) and its subsequent extension (from 2022 onward). My research has focused on computational astrophysics, with particular emphasis on dust evolution and modelling using the *Pencil Code*, which remains my principal research tool.

This project builds directly on that foundation, and I see substantial potential for advancing the state of the art through the integration of new physical processes, improved numerical treatments, and closer connections to observational constraints. The proposed work is well positioned to deliver significant progress in our understanding of dust physics in astrophysical environments.

In practical terms, my collaboration will include:

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- **Scientific guidance and mentorship:** Regular virtual meetings with the principal investigators and the prospective PhD student to support model development, numerical implementation, and the integration of simulation results with observational or theoretical frameworks.
- **Research visits and exchange:** Active participation in reciprocal visits, including hosting collaborators at Nordita and visiting the host institution, to facilitate intensive collaboration, training, and knowledge transfer.
- **Joint analysis and dissemination:** Close involvement in the interpretation of results, co-authorship of scientific publications, and contribution to the broader dissemination of outcomes within the international research community.

Through these activities, I will contribute both technical expertise and strategic input, ensuring continuity with prior developments while supporting the project's ambition to move beyond the current state of the field. Of critical relevance to this project is the access to Lumi highly parallel GPU infrastructure. I will contribute to the extension of the dust processing modules of Pencil Code updated with Paperboats functionality to the GPU based PC-A code.

Overall Assessment

This project is both timely and scientifically compelling. It addresses key open questions in computational astrophysics using a robust and well-motivated methodological framework. In my assessment, it represents a clear step forward in the field and has strong potential for high-impact scientific outcomes.

I therefore wholeheartedly support this application and look forward to collaborating closely with the team to further develop and extend this line of research.

Please do not hesitate to contact me should you require any additional information regarding my planned contribution.

Yours sincerely,

Lars Mattsson
Assistant Professor in Astrophysics

Dr. Florian Kirchschlager

Department of Physics and Astronomy

Universiteit Gent

Proeftuinstraat 86 N3, Belgium

florian.kirchschlager@ugent.be

Date: 29 April 2026

To: The Academy of Finland

Re: Letter of Collaboration for the Dust Modelling Project Application (PI: Maarit Korpi-Lagg)

To the Grant Review Committee,

I am writing to express my strong support for the funding application submitted by Maarit Korpi-Lagg and Frederick Gent regarding the dust modelling project. I am highly enthusiastic about this research and formally commit to collaborating with their team should the Academy of Finland fund the proposal.

Below are the details of my involvement, as per the Academy's guidelines for letters of collaboration.

Collaborator Information

- **Name:** Florian Kirchschlager
- **Current position:** Post-Doctoral researcher
- **Current organisation:** Ghent University (Belgium)
- **Upcoming position:** Heisenberg fellow at Hamburg University (Germany, from September 2026 on)

Purpose, Objectives, and Added Value of the Cooperation
The primary objective of my cooperation is to support the dust modelling efforts by providing specialized expertise in dust destruction simulations. I have developed the dust post-processing code Paperboats that will be used to achieve the scientific goals outlined in the proposal. From September 2026 on, I will start a new position at Hamburg University and extend the work in this field. My upcoming research group will add value to this collaboration through the integration of its established methodologies with the project team's core modelling capabilities.

Our collaboration will be practically implemented through the following activities:

- **Research Consultation:** Regular virtual meetings with the PIs and the prospective PhD student to guide model development and data integration.

- **Resource Sharing:** The provision and joint use of existing datasets.
- **Researcher Visits:** Visting the PIs and the prospective PhD student, or hosting them for a research visit at Hamburg University to train them on dust destruction simulations with Paperboats.
- **Dissemination:** Joint evaluation of the project findings and co-authorship of the resulting scientific publications.

GHENT UNIVERSITY

This dust modelling project represents a timely and highly significant scientific endeavor and I look forward to the opportunity to work alongside Maarit and Fred on this research.

Please do not hesitate to contact me should you require any further information regarding my planned contribution.

Sincerely,

Dr. Florian Kirchschlager